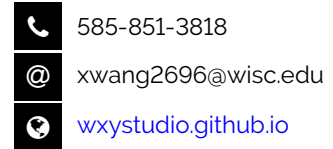


Xiaoyu Wang



RESEARCH INTERESTS

Remote sensing, crop yield prediction, knowledge-guided machine learning, AI for science

EDUCATION

2023.1 – Present	Ph.D. Student Biological Systems Engineering <i>Minor: Computer Sciences</i> <i>Courses:</i> CS 533 (Image Processing); CS 760 (Machine Learning); CS 759 (High Performance Computing for Applications in Engineering)	University of Wisconsin–Madison
2022.8 – 2023	Graduate Coursework Golisano College of Computing and Information Sciences <i>Courses:</i> CSCI 820 (Quantitative Foundations); CSCI 865 (Deep Learning)	Rochester Institute of Technology
2017 – 2021	B.S. Computer Science and Technology	Xi'an Jiaotong University

RESEARCH AND PROFESSIONAL EXPERIENCE

2021.4 – 2022.4	Research Intern Conducted research on audio front-end processing, with a focus on speech separation and enhancement.	Microsoft Research Asia, supervised by Xiangyu Kong and Xiulian Peng
2020.9 – 2021.3	Research Intern Worked on autonomous driving systems, including sensor calibration algorithms and deep learning model training.	Guohang Yan and Tao Ma
2020.6 – 2020.9	Remote Summer Intern Conducted remote research on adversarial examples and federated learning.	Nanyang Technological University, supervised by Tao Bai and Jun Zhao
2019.6 – 2020.2	Research Intern Contributed to remote sensing projects involving semantic segmentation and change detection.	INNNO
2019.1 – 2019.9	Research Assistant Conducted research on distributed generative adversarial networks.	College of Artificial Intelligence, Xi'an Jiaotong University, supervised by Jinjun Wang
2017.10 – 2018	Research Assistant Conducted research on computer security and deep learning.	Xi'an Jiaotong University, supervised by Jinsong Han

PUBLICATIONS

Journal
2025
Learning County from Pixels: Corn Yield Prediction with Attention-Weighted Multiple Instance Learning
Xiaoyu Wang, Yuchi Ma, Yijia Xu, Qunying Huang, Zhengwei Yang, Zhou Zhang
Accepted to *International Journal of Remote Sensing (IJRS)*, 2025.

Conference

2025

Knowledge-Guided Machine Learning Model with Soil Moisture for Corn Yield Prediction under Drought Conditions

Zhengwei Yang, **Xiaoyu Wang**, Jingyi Huang, Zhou Zhang

Accepted to IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2025.

2024

County-Level Crop Yield Prediction Using SMAP-Derived Data Products and Deep Learning Model

Zhengwei Yang, **Xiaoyu Wang**, Jingyi Huang, Zhou Zhang

Accepted to IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2024.

2022

Multi-Modal Multi-Correlation Learning for Audio-Visual Speech Separation

Xiaoyu Wang, Xiangyu Kong, Xiulian Peng, Yan Lu

Accepted to Interspeech, 2022.

2021

A Data-Free Approach for Targeted Universal Adversarial Perturbation

Xiaoyu Wang, Tao Bai, Jun Zhao

Accepted to SciSec, 2021.

Preprint

2025

Knowledge-Guided Machine Learning Model with Soil Moisture for Corn Yield Prediction under Drought Conditions

Xiaoyu Wang, Yijia Xu, Jingyi Huang, Zhengwei Yang, Zhou Zhang

Submitted.

GRANTS

BSE Travel Award, University of Wisconsin–Madison, 2025. (\$1,000)

AWARDS & HONORS

Third-Class Award, Xi'an Jiaotong University, 2017. (Top 20% GPA)

PROFESSIONAL SERVICE

Journal Reviewer

International Journal of Applied Earth Observation and Geoinformation (JAG)

TEACHING EXPERIENCE

2024

Teaching Assistant, BSE 405: Artificial Intelligence in Agriculture

University of Wisconsin–Madison

Prepared homework materials and Google Earth Engine lab code for course instruction.

PROGRAMMING LANGUAGES & SKILLS

Programming: Python, C++, Shell, MATLAB, CUDA C

Python Environment: Anaconda

Development: Ubuntu, Git, CMake, Docker, LaTeX, Overleaf

GPU Computing: CUDA, cuDNN, cuBLAS

TOOLS

GIS & Remote Sensing: Google Earth Engine, QGIS, ArcGIS, GDAL

Deep Learning: PyTorch, TensorFlow, TensorRT

Computer Vision: OpenCV

SLAM & Robotics: PCL, ROS

Audio Processing: librosa, Asteroid, FFmpeg, Kaldi